

SQL Project on Online Book Store

Data used:

- 1. Books.csv → Contains (Book_ID, Title, Author, Genre, Published_Year, Price, Stock)
- 2. Customers.csv → Contains (Customer_ID, Name, Email, Phone, City, Country)
- 3. Orders.csv → Contains (Order_ID, Customer_ID, Book_ID, Order_Date, Quantity, Total_Amount)

Queries:

- 1. Retrieve all books in the “Fiction” genre.
- 2. Find books published after the year 1950.
- 3. List all the customers from Canada.
- 4. Show orders placed in November 2023.
- 5. Retrieve the total stock of books available.
- 6. Find the details of the most expensive book.
- 7. Show all the customers who ordered more than 1 quantity of a book.
- 8. Retrieve all orders where the total amount exceeds ₹20.
- 9. List all genres available in the books table.
- 10. Find the book with the lowest stock.
- 11. Calculate the total revenue generated from all orders.
- 12. Retrieve the total number of books sold for each genre.
- 13. Find the average price of books in the “Fantasy” genre.
- 14. List customers who placed at least 2 orders.
- 15. Find the most frequently ordered book.
- 16. Show the top 3 most expensive books of “Fantasy” genre.
- 17. Retrieve the total quantity of books sold by each author.
- 18. List the cities where customers who spent over ₹30 are located.
- 19. Find the customer who spent the most on orders.
- 20. Calculate the remaining stock of books.

Note:- All the commands are written in upper case only.

--Creating Books Table:

```
CREATE TABLE books(Book_ID SERIAL PRIMARY KEY,
                    Title VARCHAR(100),
                    Author VARCHAR(100),
                    Genre VARCHAR(50),
                    Published_Year INT,
                    Price NUMERIC(10,2),
                    Stock INT);
```

SELECT * FROM books;

Data Output Messages Notifications							
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	book_id [PK] integer	title character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer

--Importing Books Data from .csv File:

```
COPY books (Book_ID, Title, Author, Genre, Published_Year, Price, Stock)
FROM 'K:\Books.csv'
CSV HEADER;
```

SELECT * FROM books;

Data Output Messages Notifications							
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	book_id [PK] integer	title character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer
1	1	Configurable modular throughput	Joseph Crane	Biography	1949	21.34	100
2	2	Persevering reciprocal knowledge user	Mario Moore	Fantasy	1971	35.80	19
3	3	Streamlined coherent initiative	Derrick Howard	Non-Fiction	1913	15.75	27
4	4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8

```
CREATE TABLE customers(Customer_ID SERIAL PRIMARY KEY,
                        NAME VARCHAR(100),
                        Email VARCHAR (100),
                        Phone VARCHAR(15),
                        City VARCHAR(50),
                        Country VARCHAR(150));
```

Data Output

Messages

Notifications

SQL

	customer_id [PK] integer	name character varying (100)	email character varying (100)	phone character varying (15)	city character varying (50)	country character varying (150)
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```
COPY customers (Customer_ID, Name, Email, Phone, City, Country)
FROM 'K:\Customers.csv'
CSV HEADER;
```

Data Output		Messages	Notifications			
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	customer_id [PK] integer	name character varying (100)	email character varying (100)	phone character varying (15)	city character varying (50)	country character varying (150)
1	1	Deborah Griffith	balljoseph@wright-keith.net	1234567891	South Craigfort	Denmark
2	2	Crystal Clements	kimberlybennett@curtis.com	1234567892	East Derekberg	Nicaragua
3	3	Susan Fuller	beanmichael@burnett-stewart.net	1234567893	Austinbury	Equatorial Guinea
4	4	Jamie Ramirez	amandahood@warren.com	1234567894	Dianamouth	Slovenia

```
CREATE TABLE orders(Order_ID SERIAL PRIMARY KEY,
                    Customer_ID INT REFERENCES customers(customer_id),
                    Book_ID INT REFERENCES books(book_id),
                    Order_Date DATE,
                    Quantity INT,
                    Total_Amount NUMERIC(10,2));
```

Data Output

Messages

Notifications

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SQL

	order_id [PK] integer	customer_id integer	book_id integer	order_date date	quantity integer	total_amount numeric (10,2)
--	--------------------------	------------------------	--------------------	--------------------	---------------------	--------------------------------

```
COPY orders (Order_ID, Customer_ID, Book_ID, Order_Date, Quantity, Total_Amount)
FROM 'K:\Orders.csv'
CSV HEADER;
```

Data Output		Messages	Notifications			
						
						
SQL						
	order_id [PK] integer	customer_id integer	book_id integer	order_date date	quantity integer	total_amount numeric (10,2)
1	1	84	169	2023-05-26	8	188.56
2	2	137	301	2023-01-23	10	216.60
3	3	216	261	2024-05-27	6	85.50
4	4	433	343	2023-11-25	7	301.21

--1. Retrieve all books in the “Fiction” genre.

SELECT * FROM books WHERE genre='Fiction';

Data Output Messages Notifications							
Showing rows: 1 to 60							
	book_id [PK] integer	title character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer
1	4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
2	22	Multi-layered optimizing migration	Wesley Escobar	Fiction	1908	39.23	78
3	28	Expanded analyzing portal	Lisa Coffey	Fiction	1941	37.51	79
4	29	Quality-focused multi-tasking challenge	Katrina Underwood	Fiction	1905	31.12	100
5	31	Implemented encompassing conglomerati...	Melissa Taylor	Fiction	2010	21.23	44
6	39	Optimized national process improvement	Megan Goodwin	Fiction	1978	10.99	42

--2. Find books published after the year 1950.

SELECT * FROM books WHERE published_year>1950;

Data Output Messages Notifications							
Showing rows: 1 to 292							
Page No:							
	book_id [PK] integer	title character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer
1	2	Persevering reciprocal knowledge user	Mario Moore	Fantasy	1971	35.80	19
2	4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
3	5	Adaptive 5thgeneration encoding	Juan Miller	Fantasy	1956	10.95	16
4	6	Advanced encompassing implementation	Bryan Morgan	Biography	1985	6.56	2
5	8	Persistent local encoding	Troy Cox	Science Fiction	2019	48.99	84
6	9	Optimized interactive challenge	Colin Buckley	Fantasy	1987	14.33	70

--3. List all the customers from Canada.

SELECT * FROM customers WHERE country='Canada';

Data Output Messages Notifications						
Showing rows: 1						
	customer_id [PK] integer	name character varying (100)	email character varying (100)	phone character varying (15)	city character varying (50)	country character varying (150)
1	38	Nicholas Harris	christine93@perkins.com	1234567928	Davistown	Canada
2	415	James Ramirez	robert54@hall.com	1234568305	Maxwelltown	Canada
3	468	David Hart	stokesrebecca@gmail.com	1234568358	Thompsonfurt	Canada

--4. Show orders placed in November 2023.

SELECT * FROM orders WHERE order_date BETWEEN '2023-11-01' AND '2023-11-30';

Data Output Messages Notifications						
Showing rows: 1						
	order_id [PK] integer	customer_id integer	book_id integer	order_date date	quantity integer	total_amount numeric (10,2)
1	4	433	343	2023-11-25	7	301.21
2	19	496	60	2023-11-17	9	316.26
3	75	291	375	2023-11-30	5	170.75
4	132	469	333	2023-11-22	7	194.32
5	137	474	471	2023-11-25	8	363.04
6	163	207	384	2023-11-23	3	101.76
Total rows: 25 Query complete 00:00:00.170						

--5. Retrieve the total stock of books available.

SELECT SUM(stock) AS total_stock_of_books FROM books;

Data Output Messages Notifications	
Showing rows: 1	
	total_stock_of_books bigint
1	25056

--6. Find the details of the most expensive book.

SELECT * FROM books ORDER BY price DESC LIMIT 1;

Data Output Messages Notifications							
Showing rows: 1 to 1							
	book_id [PK] integer	title character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer
1	340	Proactive system-worthy orchestration	Robert Scott	Mystery	1907	49.98	88

--7. Show all the customers who ordered more than 1 quantity of a book.

SELECT c.*, o.quantity AS quantity_ordered
FROM customers c JOIN orders o
ON c.customer_id=o.customer_id
WHERE o.quantity>1;

Data Output Messages Notifications							
Showing rows: 1 to 438 Page No: 1 of 1							
	customer_id integer	name character varying (100)	email character varying (100)	phone character varying (15)	city character varying (50)	country character varying (150)	quantity_ordered integer
1	84	Gary Blair	leonardtaylor@bowers.net	1234567974	Lake Paul	Armenia	8
2	137	Steven Miller	tsummers@yahoo.com	1234568027	North Keith	Papua New Guinea	10
3	216	Phillip Allen	brianwatkins@gmail.com	1234568106	Kelseyfort	Micronesia	6
4	433	Corey Wells	swagner@robertson.net	1234568323	East David	Zambia	7
5	14	John Wood	johnsonalexander@gmail.com	1234567904	Richardsonville	New Caledonia	7
6	439	Shane Chang	melanie56@fleming.com	1234568329	Ramosstad	Liechtenstein	5
7	195	Dominique Turner	tracy39@smith-miller.com	1234568085	Rogersborough	Djibouti	6

--8. Retrieve all orders where the total amount exceeds \$20.

SELECT * FROM orders WHERE total_amount>20 ORDER BY total_amount;

Data Output Messages Notifications						
	order_id [PK] integer	customer_id integer	book_id integer	order_date date	quantity integer	total_amount numeric (10,2)
1	307	368	133	2023-11-17	1	20.96
2	100	207	63	2023-07-14	1	22.38
3	260	112	170	2024-05-07	2	22.56
4	319	172	296	2024-06-27	2	22.98
5	345	325	253	2023-07-04	2	23.32
6	494	117	127	2024-11-23	2	23.32

--9. List all genres available in the books table.

SELECT DISTINCT (genre) FROM books;

Data Output Messages Notifications	
	genre character varying (50)
1	Romance
2	Biography
3	Mystery
4	Fantasy
5	Fiction
6	Non-Fiction
7	Science Fiction

--10. Find the book with the lowest stock.

SELECT * FROM books ORDER BY stock LIMIT 1;

Data Output Messages Notifications							
Showing rows: 1 to 1							
	book_id [PK] integer	title character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer
1	44	Networked systemic implementation	Ryan Frank	Science Fiction	1965	13.55	0

```
SELECT SUM(total_amount) AS total_revenue_generated FROM orders;
```

--12. Retrieve the total number of books sold for each genre.

	genre character varying (50)	total_books_sold bigint
1	Romance	439
2	Biography	285
3	Mystery	504
4	Fantasy	446
5	Fiction	225
6	Non-Fiction	351
7	Science Fiction	447

```
SELECT AVG(price) AS average_price_of_fantasy_genre FROM books WHERE genre='Fantasy';
```

--14. List customers who placed at least 2 orders.

Data Output	Messages	Notifications	
         			
	customer_id [PK] integer	name character varying (100)	number_of_orders_placed bigint
1	184	Richard McLaughlin	2
2	272	Carl Smith	3
3	22	Stacey Adams	3
4	173	Victoria Dixon	2
5	189	Jason Bell	2
6	117	Molly Murphy	2
7	125	Rebecca Perez	2

```
SELECT b.*, COUNT(o.book_id) AS frequency_of_order
FROM books b JOIN orders o
ON b.book_id=o.book_id
GROUP BY b.book_id
ORDER BY COUNT(o.book_id) DESC LIMIT 1;
```


Data Output Messages Notifications								
Showing rows: 1 to 1 Page No: 1 of 1								
	book_id [PK] integer	title character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer	frequency_of_order bigint
1	273	Devolved zero administration process improvement	Emily James	Romance	1929	12.43	23	4

--16. Show the top 3 most expensive books of “Fantasy” genre.

SELECT * FROM books WHERE genre='Fantasy' ORDER BY price DESC LIMIT 3;

Data Output Messages Notifications								
Showing rows: 1 to 3								
	book_id [PK] integer	title character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer	
1	240	Stand-alone content-based hub	Lisa Ellis	Fantasy	1957	49.90	41	
2	462	Innovative 3rdgeneration database	Allison Contreras	Fantasy	1988	49.23	62	
3	238	Optimized even-keeled analyzer	Sherri Griffith	Fantasy	1975	48.97	72	

--17. Retrieve the total quantity of books sold by each author.

SELECT b.book_id, b.author, SUM(o.quantity) AS total_quantity_sold
FROM books b JOIN orders o
ON b.book_id=o.book_id
GROUP BY b.book_id, b.author;

Data Output Messages Notifications			
SQL			
	book_id [PK] integer	author character varying (100)	total_quantity_sold bigint
1	184	Megan Jones	9
2	477	Chris Brown	13
3	273	Emily James	24
4	394	Karen Reese	1
5	51	Douglas Bush	3
6	272	Daniel Gillespie	10
7	70	Mary Warren	7

--18. List the cities where customers who spent over \$30 are located.

SELECT c.customer_id, c.city, SUM(o.total_amount) AS total_amount_spent
FROM customers c JOIN orders o
ON c.customer_id=o.customer_id
GROUP BY c.customer_id
HAVING SUM(o.total_amount)>30;

Data Output Messages Notifications			
SQL			
	customer_id [PK] integer	city character varying (50)	total_amount_spent numeric
1	87	East Tylertown	144.30
2	184	Bestbury	289.18
3	273	New Joehaven	321.12
4	394	Richardburgh	62.60
5	272	Lake Samantha	534.76
6	70	West Kimberlyhaven	466.60
7	176	Matthewport	141.45

-- 19. Find the customer who spent the most on orders.

SELECT c.customer_id, c.name, SUM(o.total_amount) AS total_amount_spent
FROM customers c JOIN orders o
ON c.customer_id=o.customer_id
GROUP BY c.customer_id, c.name
ORDER BY SUM(o.total_amount) DESC
LIMIT 1;

